

Matt Williams

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ML Engineer focused on LLM systems, large-scale data pipelines, and end-to-end AI product development.

SKILLS

Languages: Python (Pandas, NumPy), SQL, Java, TypeScript

Machine Learning: PyTorch, LoRA (PEFT), Unsloth, Hugging Face Transformers

Full-Stack & Infrastructure: React, Tailwind, Django, Node.js, FastAPI, Snowflake, PostgreSQL, Docker

EXPERIENCE

Crypto Asset Recovery — AI Machine Learning Engineer (February 2026 - Present)

- Improved hash recovery success rate from 30% → 35% across a 130M+ row dataset, increasing recovered credentials through optimized attack strategies
- Leveraged Hashcat with ML-driven prioritization to rank and target the top 5–10% highest-probability guesses, maximizing recovery within 10–100 guess constraints and significantly reducing brute-force search space
- Built and trained PyTorch models on 10M+ cleaned credential samples, improving top-N guess accuracy by ~12–18% over frequency-based baseline methods
- Designed large-scale data pipelines processing 130M+ records, reducing dataset size by ~25–40% via deduplication and normalization and cutting preprocessing runtime from ~48 hours to ~18–24 hours

Builda — ML-Powered Text-to-CAD System (February 2026 - Present)

- Built a Fusion 360 add-in that generates parametric CAD geometry from natural language using fine-tuned LLMs and a structured CAD DSL
- Fine-tuned and evaluated multiple LLMs using LoRA-based PEFT with PyTorch and Unsloth on NVIDIA DGX infrastructure, optimizing model performance
- Developed an end-to-end ML inference pipeline, converting model outputs into executable Fusion 360 API calls with schema-constrained validation for reliable geometry generation
- Built a model evaluation and feedback system, including a comparison add-in and human-in-the-loop (approve/reject) workflow to benchmark outputs

Business Automation — AI Engineering Intern (November 2025 - February 2026)

- Built QueryMango, a database monitoring tool tracking 10+ tables and detecting pipeline failures with real-time alerting, improving data freshness visibility
- Queried Snowflake to monitor pipeline health, identifying and resolving inconsistencies in 5–10% of datasets

STS Aerospace — Engineering Intern (May 2024 - September 2024)

- Designed Slip-On Cuffs in SolidWorks, introducing a new molding method that improved efficiency in the molding department
- Analyzed operational performance using Excel and Python to quantify efficiency gains, including cost savings and increased daily production output

EDUCATION

University of New Hampshire, Durham, NH — B.S. in Mathematics, Minor in Mechanical Engineering

CERTIFICATIONS

IBM - SQL Querying and Data Architecture Specialization (November 2025 - February 2026)

Microsoft/Coursera - Data Analysis & Visualization with Power BI (November 2025 - February 2026)

NASA - L'Space Mission Concept Academy (December 2023 - May 2024)